

Simulating Generalized Ground Multi-Agent Robotic Search Informed by Graph Neural Networks

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5/21/25

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The Problem

Search and rescue operations can often:

- Be slow and tedious
- Incur high costs in equipment, time and other resources
- Result in failure due to human error or insufficient time
 - Survival rate drops to 20% after 72 hours [1]



Figure 1. U.S. Coast Guard MH-60 Jayhawk helicopter conducting a mountain rescue. (U.S. Coast Guard, n.d.)

Background

Multi-agent robotic systems have a wide range of applications:

- Warehouse management
- Autonomous driving
- Terrain search (and rescue)

Typically relies on reinforcement learning (RL), but the RL algorithm, centralization, and communication methods can vary.



(Prorok Lab, 2021)

Other Solutions

- Using an approach based on floodfill
 - Ineffective due to multiple agents
 - More effective in structured environments like a maze
- Other Reinforcement Learning algorithms
 - Deep Q-Network
 - Requires discrete action space
 - Advantage Actor-Critic
 - Shown to have worse performance than MAPPO [4]

Why is Ours Better?

- Ground-based search allows for more thorough search than drones, yet greater speed and accuracy than human teams
- Utilizing multiple agents maximizes efficiency and decentralized communication reduces dependencies and vulnerabilities
- Multi-agent Reinforcement Learning algorithm enhanced by GNNs should improve performance
- Algorithm is scalable to any number of robots, any environment dimensions, and any number of obstacles

Impact

- Improvement in search and rescue missions that could save lives in the aftermath of a natural disaster
- Other possible applications such as locating object in inventory or organization for convenience
- Improves the current field of research of coordinating multiple robots



Methods – Reinforcement Learning

A decorative graphic in the top right corner consisting of a complex network of white lines and small circles on a purple background, resembling a circuit board or neural network connections.

- Agents interact with environment through actions
- Actions result in reward/punishment
- Actions are determined by rewards and current observations
 - Optical camera
 - LiDAR scanner
 - Shared messages
- Policy determines what action should be taken at each time step

Methods – MARL

- Multi-Agent Reinforcement Learning adds another layer of complexity
 - Cooperation vs Competition (in this case, cooperation)
- MARL encourages coordinated behaviors through
 - Interactions between agents (messages)
 - Individual vs global rewards
- Multiple different approaches for training



Methods – MAPPO

- MAPPO - Multi-Agent Proximal Policy Optimization
- Centralized training with decentralized execution
 - Each agent has individual policy
 - Shared critic
- The critic estimates average total reward for an agent
 - Used to compute advantage, determining how good an action was

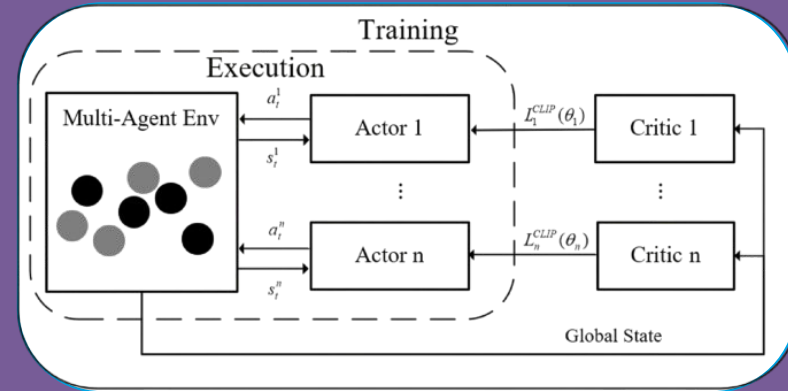


Figure 2. Diagram of MAPPO training architecture

Methods – GNNs

- A graph represents the relationship between agents
- Nodes represent the agents
- Edges represent communication links, determined by the communication range
- Node features include the observations (position, LiDAR etc.)
- GNN updates each node's features based on the neighbors
- One shared GNN policy computes actions for each agent



Methods – YOLOv11

- Fast, real-time object detection model
- Relies on CNN backbone to process image
- Predicts bounding box, objectness score, and class probabilities

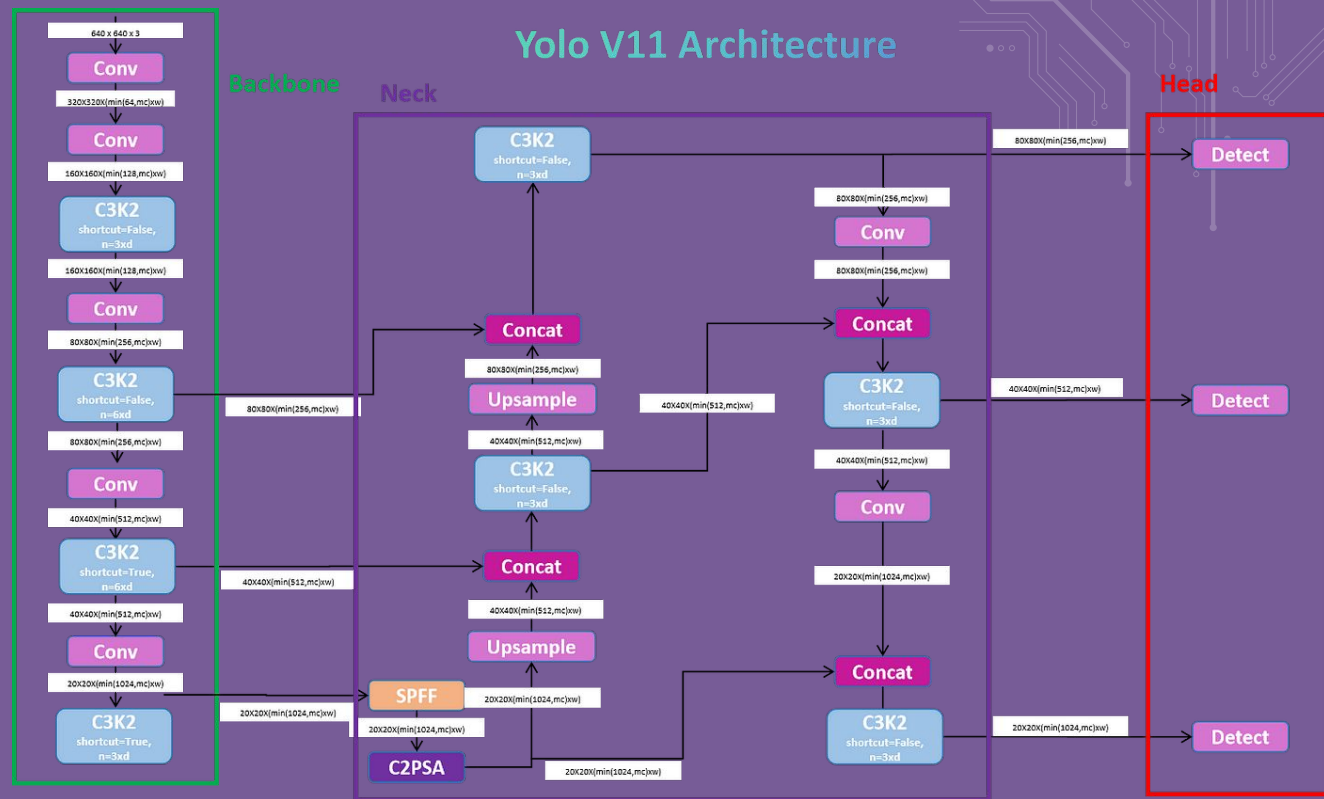


Figure 3. In-depth diagram depicting the architecture of YOLOv11

Methods – Environment Simulation

- Implemented through the Petting Zoo library
- Rendered using the Pygame library
- Treated as 2D plane
- Other parameters include
 - # of agents
 - LiDAR radius
 - Optical camera radius
 - Success range
 - # of obstacles and dimensions
 - Dimensions of the environment

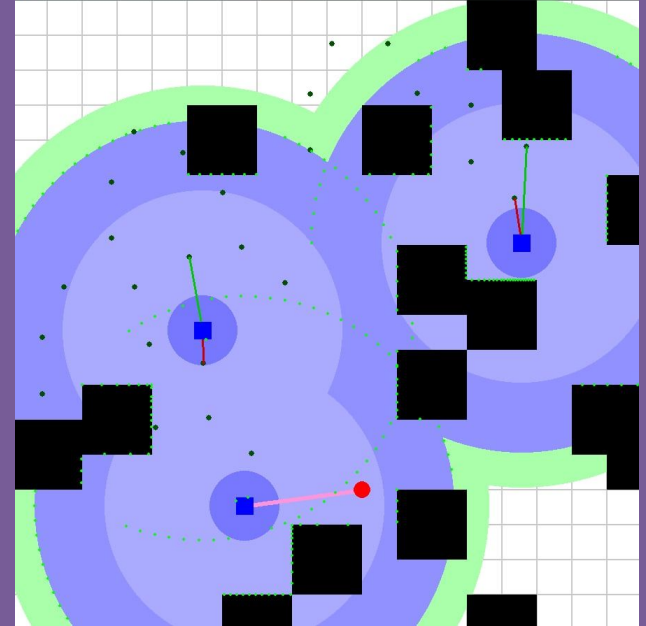


Figure 4. Example of how the environment looks after rendering

Methods – Agent (HiWonder JetAuto)

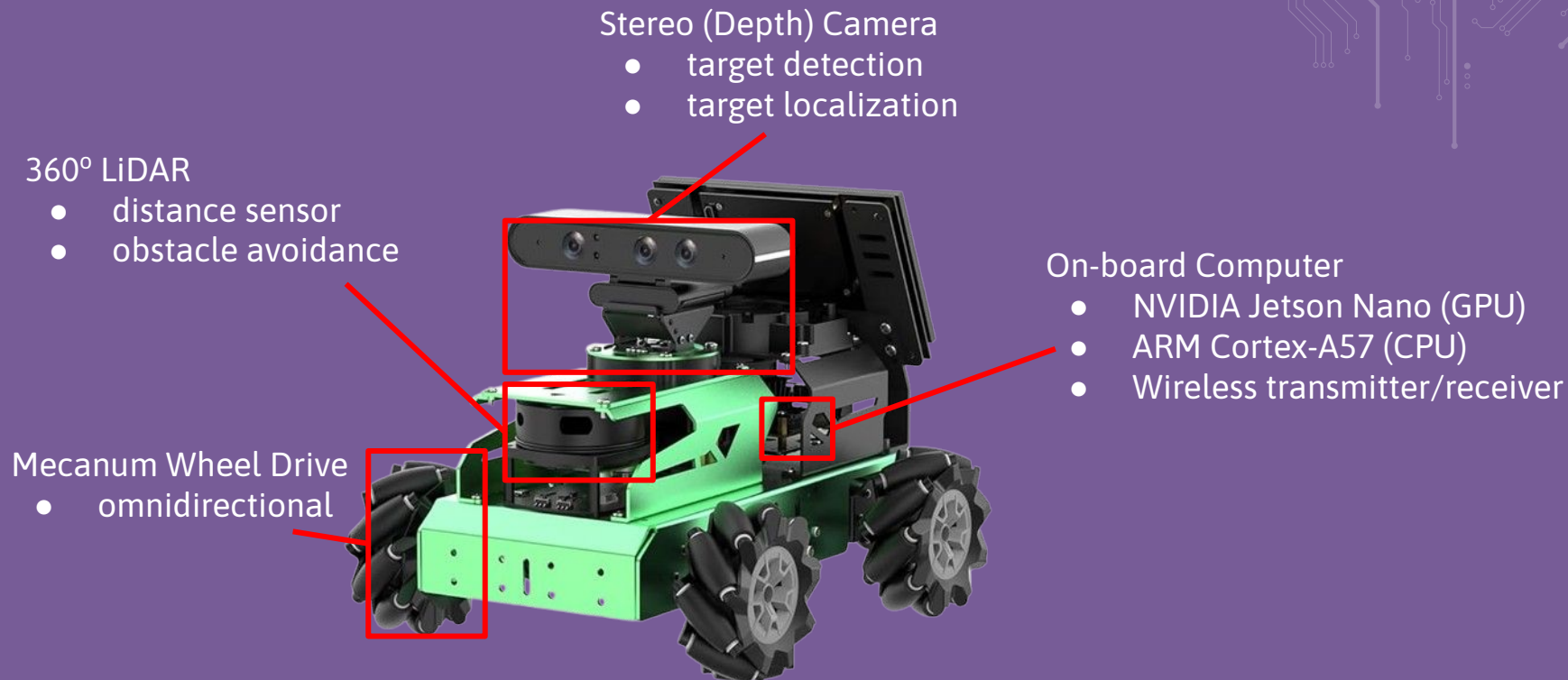
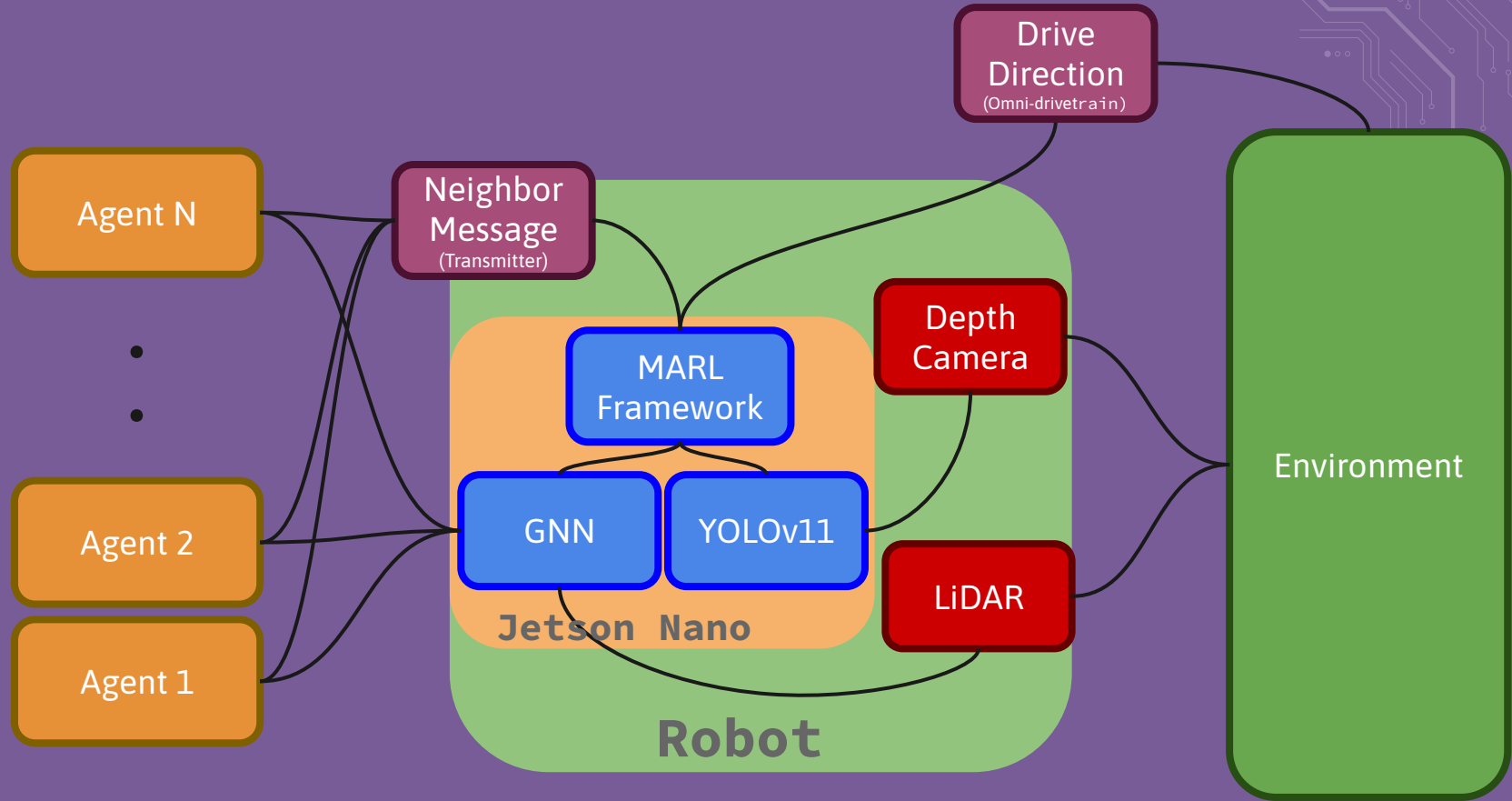
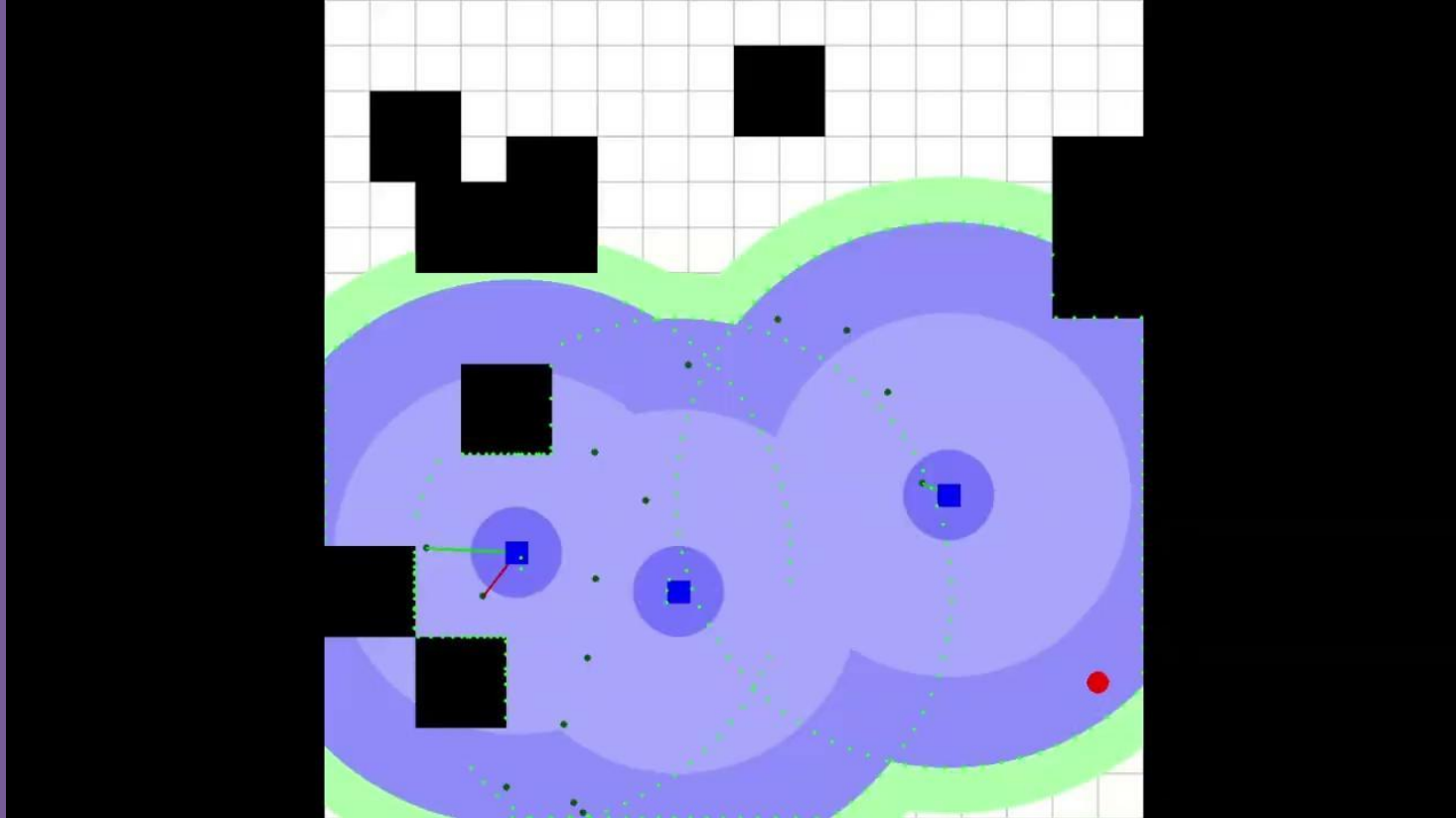


Figure 5. Capabilities of HiWonder JetAuto

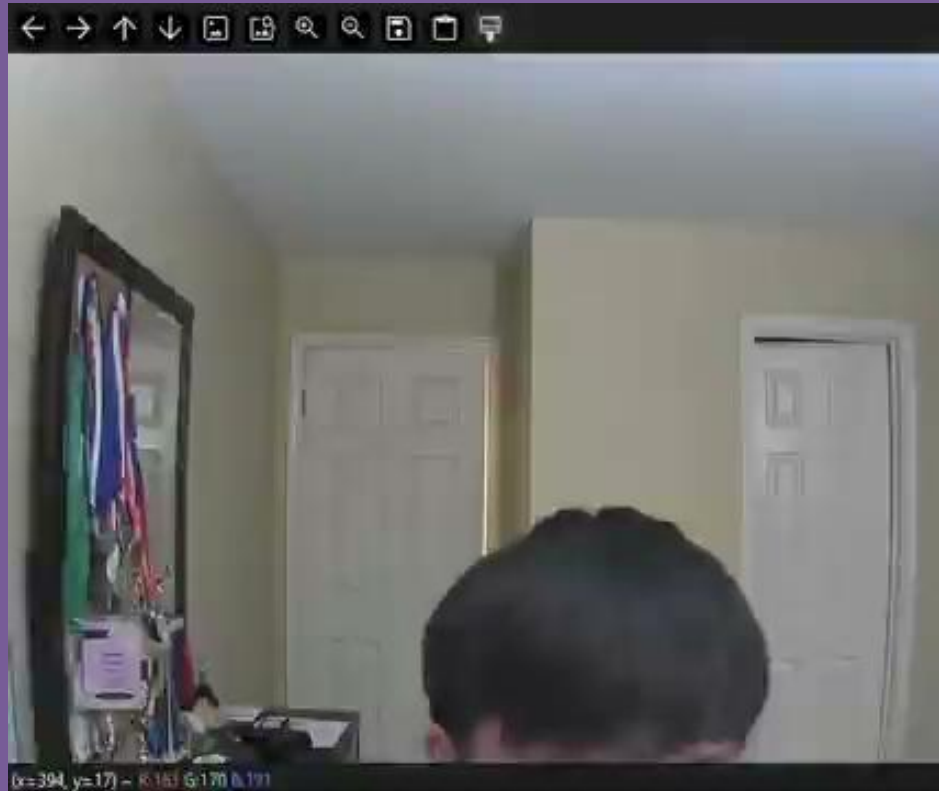
Methods – Systems Architecture



Demonstration – RL Model



Demonstration – YOLO Model



Results

- End model averages 59.4 steps per simulation, down from starting ~300 steps
- ~40% faster than a deterministic search algorithm like floodfill
- Generalizable to more agents, larger environment, and greater number of obstacles

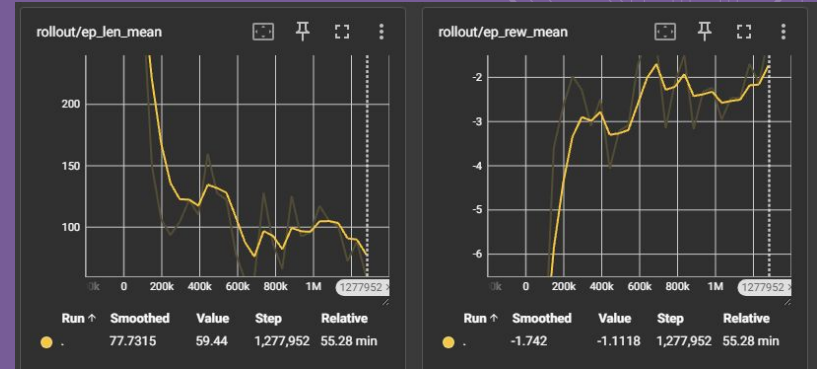


Figure 6. Graphs of the average steps and reward throughout training

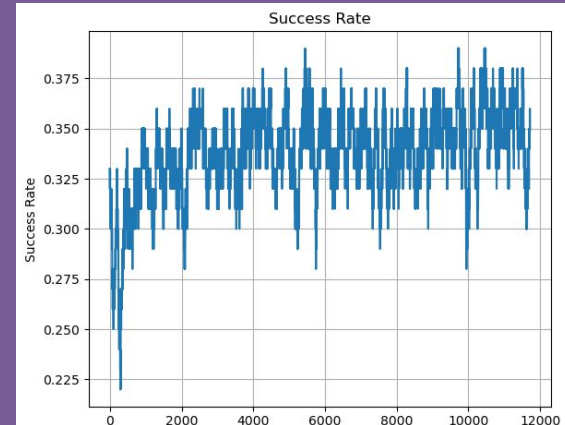
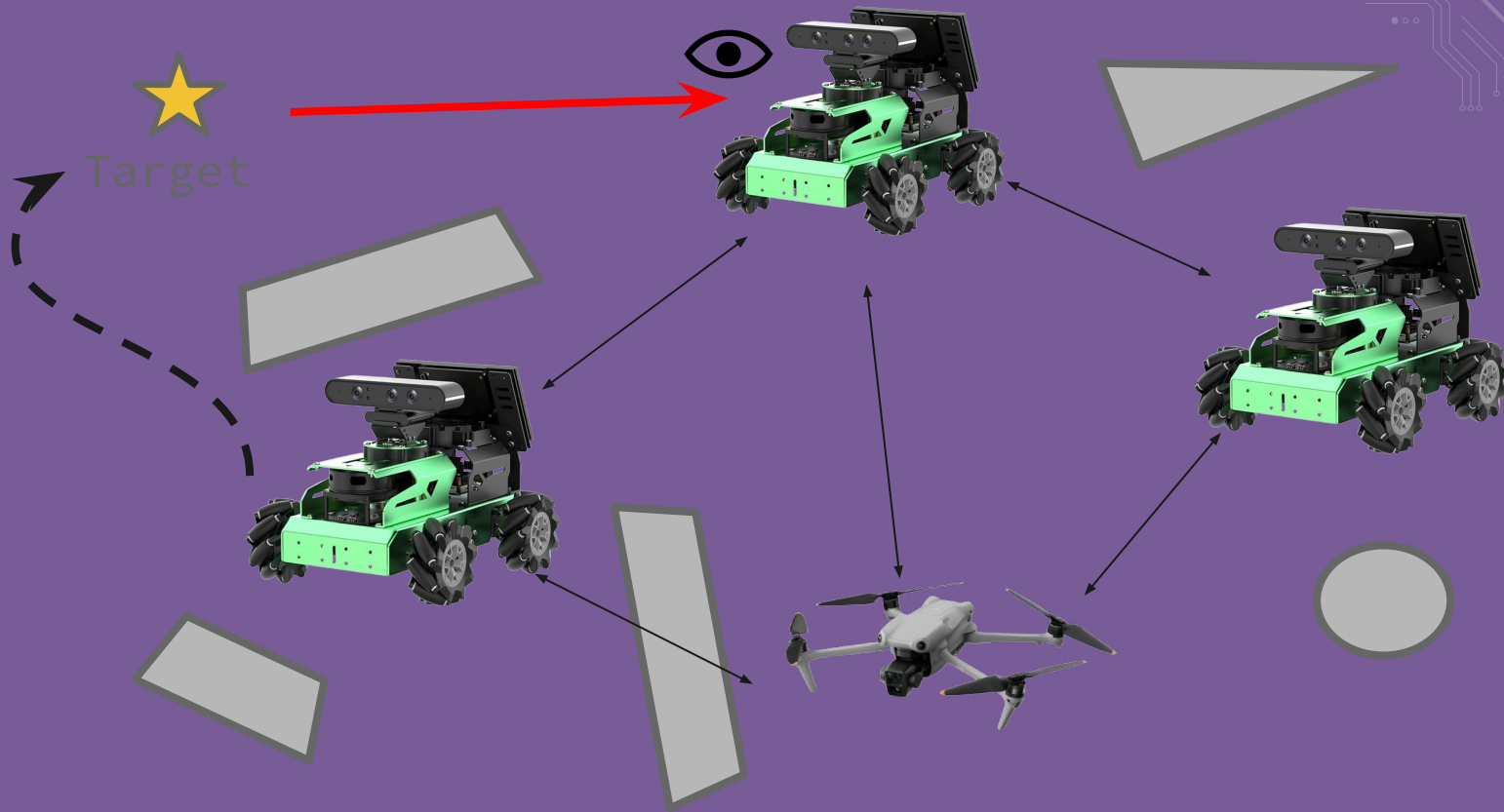


Figure 7. Graph of the success rate throughout training

Application



Challenge – Corrupted Environment

- Purchased two JetAutos
- Runs on ROS (Robot Operating Software [version Melodic]), a middleware between programs and hardware drivers. This has been implemented in Ubuntu from the manufacturer.
- To “start” the robot, roslaunch needs to be called
 - Robot A: Corruption of ROS files led us to reinstall Python and ROS, but roslaunch runs into an error when launching one of the nodes, rosout1
 - Robot B: Running roslaunch freezes after the step of “imu calibration”. We believe it came like this out of the box, and resolution requires understanding the manufacturer’s ROS system.

Limitations

- Only operating in 2-dimensional space
- Simulated environment is still not completely realistic
 - Obstacles may vary
 - Target detection might be more complicated
- Limitations on computational resources
- Unable to physically test our algorithm



Conclusion & Future Work

Successfully developed a novel Multi-Agent Reinforcement Learning algorithm for searching, while using Graph Neural Networks and simulated LiDAR scans to maximize ground-based search efficiency.

Future Ideas:

- Work on incorporating the 3rd dimension
- Utilizing optical camera outside of target detection
- Investigating Value-Decomposition Network Algorithms

References

- [1] Amini, M., Kachitvichyanukul, V., & Sirivipanich, V. (2022). Application of Soft Systems Methodology in solving disaster emergency logistics problems. *International Journal of the Analytic Hierarchy Process*, 14(1), 1–20. <https://doi.org/10.13033/ijahp.v14i1.1050>
- [2] U.S. Forest Service. (n.d.). Search and rescue team asks people to prepare for anything when headed outdoors [Photograph]. U.S. Department of Agriculture. <https://www.fs.usda.gov/about-agency/features/search-and-rescue-team-asks-people-prepare-everything-when-headed-outdoors>
- [3] Prorok Lab. (2021, November 3). A Framework for Real-World Multi-Robot Systems Running Decentralized GNN-Based Policies [Video]. YouTube. <https://www.youtube.com/watch?v=COh-WLn4iO4>

References (cont.)

[4] Abdulghani, A. M., Abdulghani, M. M., Walters, W. L., & Abed, K. H. (2024). Performance evaluation of multi-agent reinforcement learning algorithms. *Intelligent Automation & Soft Computing*, 39(2), 338–352. <https://doi.org/10.32604/iasc.2024.047017>

[5] Rao, S. N. (2024, October 22). YOLOv11 explained: Next-level object detection with enhanced speed and accuracy. Medium.
<https://medium.com/@nikhil-rao-20/yolov11-explained-next-level-object-detection-with-enhanced-speed-and-accuracy-2dbe2d376f71>



Thanks!

Any Questions?

